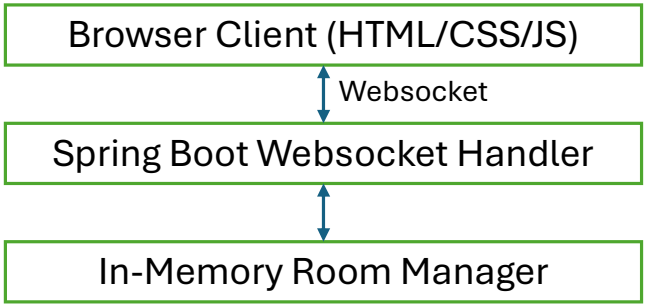
MULTIPLAYER CHAT WITH ROOMS

- I have built a real-time room based chat (similar to Discord) using Google Antigravity. This project was migrated from Python-FastAPI to Java-Spring Boot stack/framework. The Frontend has not been touched and retained as is.
- **Functionalities**
  - Room based message broadcasting to all participants presently in it.
  - System notifications for joins and departures from a room.
  - Websocket communication (Spring Boot – earlier FastAPI) between client and backend.
  - JSON Text messages with message types, usernames, server generated UTC timestamps.
  - White space stripping to avoid blank messages.
  - In-memory room membership.
  - In-memory connection management (using set and maps).

- **Simple Architecture**



What I'd Change

- When sending out messages in a particular room full of N number of people – suppose an exception is caught while sending a message, then the entire flow after it is broken and the subsequent people (websockets) in the set of people to be getting the messages will NOT get it since the exception is caught for the whole block.

- Personally, Discord has been ringing in my head since the time I read the problem statement. Since the frontend was built in two steps (one in which HTML was added and the next being decoration), I expected it to be closer to Discord's interface if AI did it.

- registry.addHandler(new ChatHandler(), "/ws/*/*").setAllowedOrigins("*"); This line allows all kinds of regex, I feel it is possible to restrict this by changing to only certain allowed regular expressions for the purpose of security.